PHARMACOVIGILANCE INTERNSHIP

ASSIGNMENT 03: CAUSALITY ASSESSMENT OF ADVERSE DRUG REACTIONS

Submitted by: Anisha Hussain

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1. OBJECTIVES

The primary objective of this assignment is to understand and apply standard causality assessment methods used in pharmacovigilance to evaluate the relationship between a suspected drug and an adverse event.

The specific objectives are:

- To understand the concept and importance of causality assessment.
- To identify drug-related, event-related, patient-related and clinical factors.
- To apply the Naranjo Algorithm to a clinical case.
- To classify suspected ADRs using the WHO-UMC causality assessment system.
- To understand implied causality in spontaneous safety reports.
- To differentiate between adverse events and adverse drug reactions.

2. INTRODUCTION

Causality assessment, also known as relatedness assessment, is the systematic process of determining the causal relationship between drug exposure and an adverse event. An adverse event occurring after administration of a medicinal product does not automatically establish that the medicine caused the event. Causality assessment helps determine whether a suspected reaction is related to the medicinal product or whether other factors may have contributed to its occurrence. In pharmaceutical companies, this activity is generally undertaken by a medical reviewer or safety physician.

The two commonly used methods are:

- Naranjo Adverse Drug Reaction Probability Scale.
- WHO-UMC Causality Assessment System.

3. FACTORS ASSOCIATED WITH CAUSALITY ASSESSMENT

Causality assessment involves four major categories of factors.

A. Drug-Related Factors

These factors concern the suspected medicinal product.

- Labeledness: Whether the reaction is already documented in the product information.
- Predisposing factors associated with the medicine.
- Concomitant medications that may interact with the suspected drug.
- Known pharmacological properties of the drug.

B. Event-Related Factors

These concern the characteristics of the adverse event.

- Nature and type of reaction.
- Clinical presentation.
- Whether the event is a recognized drug reaction.
- Whether the event is consistent with the known safety profile.

C. Patient-Related Factors

Individual patient characteristics may influence the development of adverse reactions.

- Medical history.
- Previous allergic reactions.
- Genetic predisposition and pharmacogenomic factors.
- Existing diseases and other predisposing conditions.

For example, a patient with a previous penicillin allergy may have an increased risk of developing an allergic reaction when exposed to certain related antibiotics.

D. Clinical Factors

These factors help establish the clinical relationship between drug exposure and the event.

- Temporal relationship between drug administration and event onset.
- Dechallenge: Improvement after withdrawal or reduction of the suspected drug.
- Rechallenge: Recurrence of the reaction following re-administration.
- Response to corrective treatment.

4. CAUSALITY ASSESSMENT METHODS

4.1 Naranjo Algorithm

The Naranjo Algorithm is a standardized questionnaire-based method developed by Naranjo and colleagues in 1981. It consists of ten questions designed to evaluate the probability that a suspected drug caused an adverse reaction. Each question receives a numerical score based on the response.

Interpretation of total scores:

TOTAL SCORE	CLASSIFICATION
9 or above	Definite
5 - 8	Probable
1 - 4	Possible
0 or below	Doubtful

4.2 WHO-UMC Causality Assessment System

The WHO-UMC system was developed by the World Health Organization in collaboration with the Uppsala Monitoring Centre.

It classifies suspected drug-event relationships into six categories.

CATEGORY	DESCRIPTION
Certain	A strong and clinically convincing relationship exists.
Probable/Likely	A reasonable time relationship exists, and alternative explanations are unlikely.
Possible	| A reasonable time relationship exists, but other explanations may also be responsible.
Unlikely	The timing makes a causal relationship improbable, and other explanations are more likely.
Conditional	More information is required for assessment.
Unclassifiable	The available information is insufficient or contradictory.

5. CASE STUDY 1: AMOXICILLIN-ASSOCIATED SKIN RASH

Case source:Clinical example discussed in the VirtualWorks causality assessment tutorial.

Patient and Medication Details

- Patient name: Raj (as given in the tutorial).
- Suspected drug: Amoxicillin.
- Indication: Throat infection.
- Initial dosage: 500 mg three times daily.
- Route: Oral.

Clinical Timeline

- 1 June 2021: Amoxicillin 500 mg three times daily initiated.
- 2 June 2021: Patient developed itchy skin rashes.
- 3 June 2021: Dose reduced to 250 mg three times daily.
- 4 June 2021: Skin rash disappeared.
- 5 June 2021: Dose increased again to 500 mg three times daily.
- 6 June 2021: Generalized itchy rashes reappeared.

Clinical Observation

The patient developed skin rashes following the initiation of amoxicillin treatment.
The reaction disappeared after dose reduction and reappeared following an increase in dosage.
This sequence suggests a possible dose-related association between amoxicillin exposure and the skin reaction.

5.1 Naranjo Algorithm Assessment

NO.	ASSESSMENT QUESTION	RESPONSE	SCORE
1	Are there previous conclusive reports of this reaction?	Yes	+1
2	Did the adverse event appear after administration of the suspected drug?	Yes	+2
3	Did the reaction improve after withdrawal or reduction of the drug?	Yes	+1
4	Did the reaction reappear following re-administration?	Yes	+2
5	Are there alternative causes that could have caused the reaction?	Not Known	0
6	Did the reaction reappear after placebo administration?	Not Known	0
7	Was the drug detected in toxic concentrations in body fluids?	Not Known	0
8	Was the reaction more severe when the dose was increased or less severe when decreased?	Yes	+1
9	Did the patient experience a similar reaction to the same or similar drugs previously?	Not Known	0
10	Was the adverse event confirmed by objective evidence?	Not Known	0

Total Naranjo Score: 1 + 2 + 1 + 2 + 0 + 0 + 0 + 1 + 0 + 0 = **7**

Final classification: **PROBABLE**

Interpretation

- The calculated score of 7 indicates a probable relationship between amoxicillin administration and the development of the skin rash.
- The positive temporal relationship, improvement following dose reduction and recurrence after dose escalation provide supporting evidence.
- However, the possibility of alternative causes and the absence of additional clinical information limit the certainty of the assessment.

Scoring note: The dose reduction is treated as a positive response to reduced exposure, following the tutorial's example. This is an educational application of the algorithm.

5.2 WHO-UMC Assessment

The following factors were evaluated:

CRITERIA	OBSERVATION
Temporal relationship	Reaction occurred after starting amoxicillin.
Known reaction	Skin rashes are recognized with amoxicillin.
Dechallenge	Rash disappeared after dose reduction.
Rechallenge	Rash reappeared following dose escalation.
Alternative causes	Not completely excluded.

WHO-UMC Classification: **PROBABLE/LIKELY**

Justification: The reaction demonstrated a reasonable temporal relationship with drug exposure. Improvement following dose reduction and recurrence following dose escalation strongly support a drug-related association.

However, because alternative causes have not been completely excluded and the available information is limited, the conservative classification is Probable/Likely rather than Certain.

6. ADDITIONAL CASE-BASED CLASSIFICATION

The following are hypothetical educational scenarios designed to demonstrate different WHO-UMC classifications.

- **Case 2: Certain**

Clinical scenario: A patient receives a particular medication and develops a documented immediate hypersensitivity reaction. The reaction has a well-established association with the drug, no reasonable alternative cause is identified, and an accidental re-exposure results in recurrence of the same reaction.

Assessment:

- Strong temporal relationship.
- Recognized drug reaction.
- No plausible alternative explanation.
- Recurrence after re-exposure.

WHO-UMC Classification: **CERTAIN**

Justification: The available clinical evidence strongly supports a causal relationship between the medicinal product and the reaction.

- **Case 3: Possible**

Clinical scenario: A patient receiving an antibiotic develops nausea two days after treatment initiation. However, the patient also has an underlying gastrointestinal infection that may independently explain the symptoms. Information regarding improvement after drug withdrawal is unavailable.

Assessment:

- Reasonable temporal relationship.
- Reaction may be associated with the drug.
- Underlying disease provides an alternative explanation.
- Dechallenge information is unavailable.

WHO-UMC Classification: **POSSIBLE**

Justification: A relationship with the suspected drug is plausible, but the underlying illness may also explain the event.

7. IMPLIED CAUSALITY IN SPONTANEOUS REPORTING

Implied causality refers to the assumption of a possible causal relationship when an adverse event is reported spontaneously by a patient or healthcare professional.

Spontaneous reports are generally unsolicited reports submitted to pharmaceutical companies or pharmacovigilance centres.

Even when the causal relationship is uncertain, a spontaneously reported event should not be dismissed solely because causality has not been established.

The report should be appropriately documented and evaluated according to pharmacovigilance reporting requirements.

This is important because prematurely excluding a suspected association may result in the loss of potentially valuable drug safety information.

8. DIFFERENCE BETWEEN ADVERSE EVENT AND ADVERSE DRUG REACTION

PARAMETER	ADVERSE EVENT (AE)	ADVERSE DRUG REACTION (ADR)
DEFINITION	Any untoward medical occurrence following drug exposure.	A harmful and unintended response for which a causal relationship with the drug is at least reasonably suspected.
CASUALITY	Not necessarily established.	A suspected causal relationship exists.
DRUG RELATIONSHIP	May or may not be related.	Relatedness is suspected.
EXAMPLE	Headache occurring during antibiotic treatment without established drug association.	Suspected drug-induced allergic rash following antibiotic administration.

Important distinction: *An adverse event is not automatically an adverse drug reaction. The suspected causal relationship is what distinguishes the two.*

9. VACCINE-RELATED CAUSALITY ASSESSMENT

Adverse events following immunization (AEFI) require a specific causality assessment approach.

An AEFI may be associated with:

1. Vaccine product-related reactions.
2. Vaccine quality defects.
3. Immunization error-related reactions.
4. Immunization anxiety-related reactions.
5. Coincidental events unrelated to vaccination.

The WHO approach considers five major principles:

- Strength of association.
- Consistency of association.
- Specificity of the adverse event.
- Temporal relationship.
- Biological plausibility.

Four-Step AEFI Causality Assessment Process

1. Eligibility assessment: Determine whether the reported event is suitable for causality assessment.
2. Checklist: Collect and evaluate relevant clinical and vaccination information.
3. Algorithm: Assess the available evidence using the recommended decision-making process.
4. Classification: Determine the final causality category.

The final classification may be:

- A: Consistent causal association to immunization.
- B: Indeterminate.
- C: Inconsistent causal association to immunization.
- Unclassifiable: Insufficient information for assessment.

10. ROLE OF CAUSALITY ASSESSMENT IN PHARMACOVIGILANCE

Causality assessment plays an important role in pharmaceutical drug safety operations.

Its major applications include:

- Evaluation of suspected adverse drug reactions.
- Medical review of individual case safety reports.
- Identification and evaluation of potential safety signals.

- Supporting benefit-risk assessment.
- Improving the quality of drug safety documentation.
- Supporting regulatory pharmacovigilance activities.

Causality assessment may also be supported by automated systems and machine-learning technologies, although clinical judgement and medical review remain important.

11. CONCLUSION

Causality assessment is an essential component of pharmacovigilance that helps determine the likelihood of a relationship between medicinal products and suspected adverse events.

Through this assignment, the Naranjo Algorithm and WHO-UMC causality assessment system were studied and applied to clinical scenarios.

The amoxicillin case demonstrated how temporal association, dechallenge, rechallenge and dose-response relationships contribute to the evaluation of suspected adverse drug reactions.

The exercise also highlighted the importance of patient history, alternative explanations, spontaneous reporting and accurate safety documentation.

A systematic approach to causality assessment supports effective drug safety monitoring and helps protect patient health.

12. REFERENCES

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